Mohit Srivastav

Curriculum Vitae

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MohitS704
Ph.D. Candidate at Johns Hopkins University

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Education

2022

Ph.D. Candidate, Johns Hopkins University (JHU)
Working on the Compact Muon Solenoid (CMS) experiment at the Large Hadron Collider (LHC)
Advisor: Professor Andrei Gritsan

2018–22

B.S. Physics, B.A. Computer Science, University of Virginia (UVA)
Highest Distinction in Physics, High Distinction in Computer Science

Selected Publications

CMS Collaboration. Determination of the spin and parity of all-charm tetraquarks. *Nature*, 648(8092):58–63, December 2025.

CMS Collaboration. Measurement of the higgs boson mass and width using the four-lepton final state in proton-proton collisions at $\sqrt{s} = 13$ TeV. *Physical Review D*, 111(9), May 2025.

CMS Collaboration. Off-shell higgs boson measurements: Yukawa couplings, self-coupling, compositeness, and width, 2026.

The CMS Collaboration. Combined measurements and interpretations of higgs boson production and decay in proton-proton collisions at $\sqrt{s} = 13\,\mathrm{TeV}$. *Reports on Progress in Physics*, 89(8):087801, aug 2026.

Jeffrey Davis, Andrei V. Gritsan, Lucas S. Mandacaru Guerra, Lucas Kang, Michalis Panagiotou, Jeffrey Roskes, and Mohit Srivastav. Maximizing returns: Optimizing experimental observables at the lhc. *arXiv*, 2026.

Analysis Work

2023

Off-shell Higgs Boson, *CMS Joint 4-Lepton Study Team (CJLST)*
Expert in off-shell Higgs boson analyses on CMS, working towards a full SM EFT interpretation of the off-shell Higgs boson

- Helped produce leading constraints on the width of the Higgs Boson including anomalous terms in gluon-fusion production on CMS [2]
- Collaborated to include an off-shell Higgs analysis into the CMS Higgs Grand Combination for the first time by converting Ref. [2] into the κ framework [4]
- Produced world-leading constraints on light-quark Yukawa couplings, the first direct test of Higgs boson compositeness, and experimental discovery of off-shell Higgs boson production [3]
- Helped to craft and utilize machine learning methods, as well as a novel, optimal, way to merge bins together to ensure sensitivity in a multidimensional phasespace [5]

2022–25

Spin-Parity Analysis of a Tetraquark Candidate, *CMS B-Physics*

- Found evidence for the spin and parity of 3 low-mass resonances which constitute a candidate for a tetraquark [1]
- Utilized techniques pioneered in the Higgs sector and developed them for use in B-physics

2020–22

Optimizing Cuts for Experimental Dead-time, *Mu2e Collaboration*

- Utilized machine learning to minimize experimental deadtime at high beam intensities, and studied the aging of test-stand di-counters for particle detection culminating in a Thesis in Computer Science

Technical Work

2025

Lead Developer/Maintainer, *MiLoMerge Package*

- Developer/maintainer for the MiLoMerge, package, designed to merge bins optimally to reduce separability loss between N distributions [5].
- Writing documentation and unit-tests for MiLoMerge in-line with Scikit-HEP requirements
- Version 1.0.0 released on January 15, 2026

2025

Monte Carlo Simulation Corrections, *CMS 4-lepton channel*

- Calculated simulation k-factors for the dominant $q\bar{q} \to 4\ell$ background process and $gg \to 4\ell$ Higgs and diboson production processes for the CMS experiment and applied them to real data, as presented in the LHC Higgs Working Group
- Worked in conjunction with experts at Fermilab to establish better parton showering for off-shell Higgs Boson simulation to better match data
- Produced a better filter for the 4-lepton channel to be used in off-shell Higgs Boson simulation

2022

JHUGen/JHUGen-MELA Development, *Johns Hopkins University*

- Developer for the JHUGen Package, which includes a Monte Carlo generator as well as a reweighting scheme for experimental high-energy physics, with the latest version (v7.6.0) released Jul. 31, 2026
- Created new Python bindings using PyBind11 to release a version of JHUGen-MELA in Python
- Writing documentation for the JHUGen-MELA package utilizing Doxygen
- Introducing a new set of matrix-element calculations into JHUGen-MELA through the utilization of Madgraph matrix-elements for a range of different processes.

2022

CMS Tracker Alignment, *Tracker Alignment Group*

- Utilize survey constraint data to provide initial constraints for the High-Luminosity LHC tracker upgrade alongside teams at Cornell and Fermilab
- Provide alignments for the CMS tracker and validate performance over time

Public Talks & Posters

2026

From Peak to Tail: Probing the Higgs Boson Width and New Physics with Off-Shell Four-Lepton Production
Meeting of the Division of Particles and Fields – Fermi National Accelerator Laboratory, IL

2026

Binning Bonanza: Metric Design for Quantum Observables and Efficient Dimensionality Reduction
Pheno 2026 – Pittsburgh, PA

2026

Status of k-factors for Run-3 analyses
LHC Higgs Cross Section Working Group – LHC Higgs Working Group

2025

Maximizing Impact: The Quest for Optimal Observables in Experiments
JHU Experimental Particle Physics Seminar – Johns Hopkins University, MD

2025

Blasting Off(-shell): $H^* \to ZZ \to 4\ell$ for Precision Higgs boson Measurements at CMS
Fermilab LHC Physics Center Forum – Fermi National Accelerator Laboratory, IL

2025 **Measurement of the Higgs boson width using the ZZ final state**
Fermilab Annual Users and Affiliates Meeting – Fermi National Accelerator Laboratory, IL

2025 **JHUGen-MELA Tutorial for Higgs SM EFT**
LPC EFT Workshop – Fermi National Accelerator Laboratory, IL

2024 **JHUGen-MELA Tutorial for Higgs SM EFT**
LPC EFT Workshop – University of Notre Dame, IN

2024 **JHUGen-MELA Tutorial for Off-shell Higgs Study**
LPC Off-shell Higgs Boson Workshop – Fermi National Accelerator Laboratory, IL

2023 **Analysis of Hadronic Resonance Structures in the $J/\psi J/\psi$ Invariant Mass Spectrum**
American Physical Society April Meeting – Minneapolis, MN

2022 **Improvements to Cosmic Muon Identification Using Machine Learning**
Virginia Space Grant Consortium Research Symposium – Norfolk, VA

2021 **Improvements to Cosmic Muon Identification Using Machine Learning**
Meeting of the Division of Particles and Fields – Florida State University, FL

Awards & Honors

2026 **Best overall poster presentation (Jun. 2026)**
US CMS meeting at the University of Maryland

2026 **Paul G. Agnew Teaching Award (May 2026)**
Department teaching award for continued excellence in teaching

2026 **Krieger School of Arts and Sciences Excellence in Teaching Award (Apr. 2026)**
College-wide teaching award for excellence in teaching by a graduate student

2026 **Phenomenology 2026 Symposium Student Travel Award (April 2026)**
Conference travel award to go to the Phenomenology 2026 Symposium

2025 **LPC Guest & Visitor Award (May 2025)**
Awarded monetary support to be at Fermilab to conduct central shifts at the LHC Physics Center (LPC) and assist with metrology for the outer tracker upgrade

2025 **E.J. Rhee Flair in Teaching Award (May 2025)**
Department teaching award for "creative flair" in teaching

2025 **Kupperman Family Department Travel Grant (May 2025)**
Department award to facilitate travel to summer conferences

2025 **Universities Research Association Visiting Scholars Program (Mar. 2025)**
Universities Research Association (URA) award to stat at Fermilab for June/July 2025

2024 **CMS IRIS-HEP Analysis Software Training Travel Grant (Mar. 2024)**
Awarded monetary support to travel to Princeton university to attend the IRIS-HEP software training as well as the US CMS meeting

2023 **URA P5 Fermilab Travel Grant (Dec. 2023)**
Awarded monetary support to travel to the P5 policy plan announcement town hall at Fermilab

2021 **Mitchell Undergraduate Research Scholarship (Jun. 2021)**
Awarded a scholarship to conduct research at the University of Virginia under Professor E. Craig Dukes for the Mu2e experiment

2021 **Virginia Space Grant Consortium Research Scholarship (Apr. 2021)**
Awarded a scholarship for conducting research into improving dead-time for the Mu2e experiment

SULI Argonne Lab (Jan. 2021)
Offer extended; did not accept

DAAD Rise Research Fellowship (Mar. 2020)
Offer extended; cancelled due to COVID-19

SULI Jefferson Lab (Jan. 2020)
Offer extended; cancelled due to COVID-19

Selected Teaching Experience

Head TA, *Classical Mechanics for Physics Majors*, Fall 2024, 2025, 2026
Head TA for Physics 1 for Physics Majors

TA, *General Physics Lab*, Fall 2023 – Spring 2024
TA for General Physics Lab 1 & 2

TA, *Computer Algorithms*, Summer 2020 – Spring 2021
TA for CS 4102

TA/Head TA, *Introductory Computer Science in Python*, Spring 2019 – Spring 2022
TA and Head TA for CS 1110/1111

Physics Outreach

Johns Hopkins Physics Fair Volunteer (April 2026)
Part of the LHC Exhibit talking showcasing the cloud chamber and muon detector

Fairfax Science and Engineering Fair Category Judge (March 2026)
Judging high school students at the regional science fair in Fairfax, Virginia

Johns Hopkins Physics Fair Volunteer (April 2025)
Performed the "Ping-pong cannon" demo for local families and talked about shockwaves/air pressure

Workshop/Conference Organizing and Professional Development

Science Policy and Advocacy for Research Competition (SPARC), Fall 2026
Accepted member of the SPARC competition organized by the Universities Research Association (URA), to assist "early career scientists to help bridge the gap between science and the public".

LPC EFT Workshop Local Organizing Committee
On the local organizing committee for the LPC EFT workshop held at Johns Hopkins

Other Experience

Fitness Monitor, *Ralph O'Connor Rec Center*, JHU
Helped manage facilities, play music, and interact/help patrons at the gym

SAT Tutor, *The Answer Class*, Maryland
SAT Tutor leading prep courses in various classes at high schools in the Baltimore area

Writer, *The Yellow Journal*, UVA
Collaboratively wrote articles and headlines for the Yellow Journal, the University of Virginia's satirical publication

Treasurer, *International Relations Organization*, UVA
Managed taxes and finances on the order of $30k for the 501c3 International Relations Organization at the University of Virginia